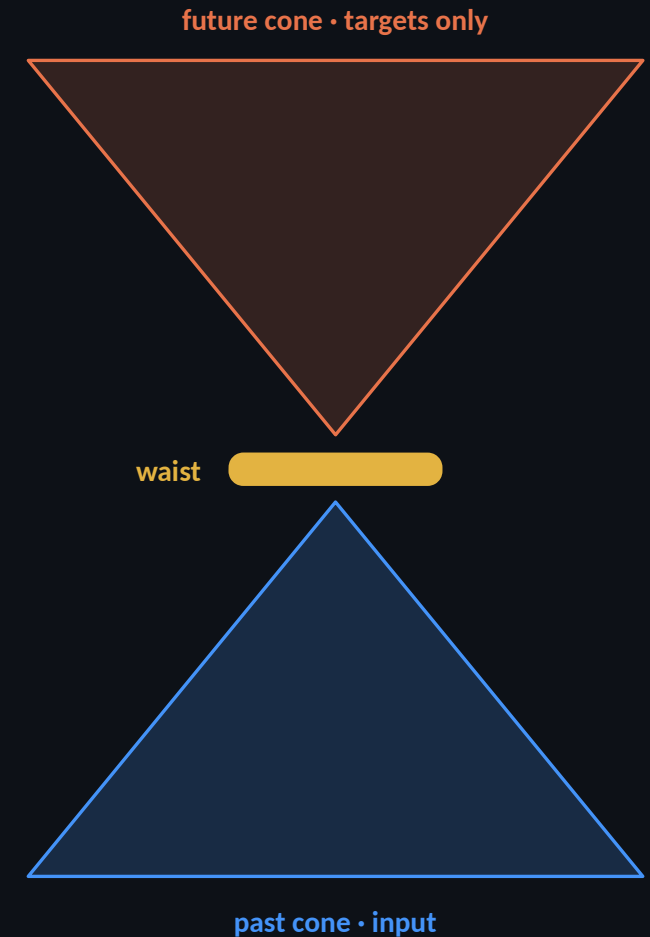


The cut-off mirrored double cone

Stage 1 as an hourglass — and a cone the model shapes itself

E-080 • design proposal • 15 September 2026 • **nothing here is measured yet**

The deck turns Chris's four points into a specification: a waist instead of a tip, a mirrored cone of targets, a random mix of four tasks, and a cone whose shape the model learns per region.



One cone, five seeds, one clear reading

E-069 • the cone-native codec

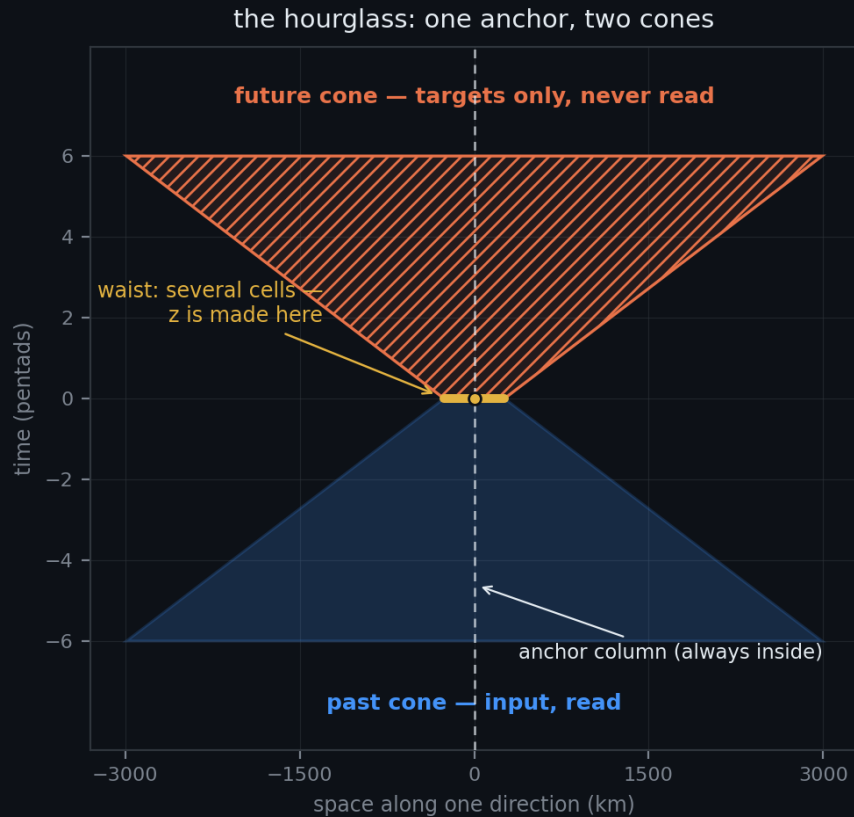
- 30 days of dots around a North Atlantic cell, one 3×3 patch at the present, targets at +1 and +2 pentads
- History did **not** put the cell's own velocity into the embedding — five seeds, two objectives, refuted
- History **did** buy persistent fields (hidden sea-surface height at ~0.36 of the bar vs 0.75–0.90) and a 7–8 % better one- and two-pentad forecast

E-071 • cone v2

- Global, harmonic climatology, profile tokens — the geometry's known faults, fixed
- Reach from the fastest mechanism × 1.5 — a hemisphere by lag 3, so dots go log-radial

The open question v2 left: a cone wide enough to never exclude a driver samples the near field at one dot per thousands of kilometres.

Past cone → waist → future cone



- **Past cone** (lags $-6 \dots -1$): read as input, dots at the per-lag ellipse plus the anchor column
- **Waist** (lag 0): a disc of several cells — the cut-off tip — read as input and reconstructed; the embedding z summarises the waist
- **Future cone** (leads $+1 \dots +6$): never read; sampled the same way and used as targets only
- **Point-mirrored**: the future cone at lead $+\ell$ is the past cone at lag $-\ell$ reflected through the anchor

A gradient does not exist within a single cell

- The waist is a disc of radius r_0 around the anchor, sampled as a small sunflower: ~13 cells at the default (one centre + 12), never fewer than the 3×3 patch's nine
- Read as input at lag 0 (values + observed flags) and reconstructed as the anchor family's target
- **What the waist buys:** thermal wind is a density gradient; geostrophy is a sea-surface-height slope — both need neighbours at the same instant, not in the past
- The waist has the same shape parameters as the cone at lag 0 (orientation, two semi-axes), so a boundary current can have an elongated waist along the flow
- The embedding z summarises the waist; one z per anchor, as today

≈ 13

cells at the waist

one centre + 12, never fewer than the nine of the 3×3 patch

1 z

per anchor, as today

so stage 2 and every archived probe see the same kind of object

Sampled like the past cone, read by nobody

+1 ... +6 pentads

30 days ahead, matching the 30 days of history

targets only

nothing in the future cone is ever read by the encoder

frozen at evaluation

the fixed v2 hourglass scores every arm and every seed

- Leads +1...+6 pentads (30 days ahead), the mirror image of the past cone through the anchor: centre $c(+\ell) = -c(-\ell)$, same axes and orientation at the same $|\ell|$
- Dots are **targets only**: each is a decoder query (channel, Δx , Δy , lead, depth) $\rightarrow (\mu, \log \sigma^2)$, scored with the Gaussian negative log-likelihood already in `cone_codec.py`
- The anchor column at every lead is always among the targets — the fixed-frame forecast stage 2 needs
- Per-lead and per-family loss, each reported against its own predict-the-mean bar (the E-069b rule)
- Evaluation geometry is **frozen** (the fixed v2 hourglass), whatever the training cone learns — so every number stays comparable across arms and seeds

One model, four questions, drawn at random per example

task	input	targets	share	what it teaches
T1 forecast	waist + past cone	future cone	0.35	the full problem
T2 snapshot forecast	waist only (past cone fully hidden)	future cone	0.15	the built-in twin: what history buys, inside one model
T3 nowcast from history	past cone under a dropout pattern; waist hidden	waist	0.25	the present from the past — persistence, advection, memory
T4 fill-in	waist + past cone, both under a dropout pattern	the hidden waist cells and dots	0.25	E-069b's reconstruction family, kept

Dropout patterns for T3 and T4, drawn independently (each fires with its own probability)

channel drop

at the waist — lag 0 only,
chan_drop_scope lag0

recent-lag band

hide every dot at $\text{lag} \leq \ell_o, \ell_o \in \{1, 2, 3\}$

bearing wedge

hide a 90° sector

random dots

each dot hidden with $p = 0.3$

whole waist

T3 only

Three rules, all learned the hard way in E-069

1 • Knowable targets only

A target channel must be visible somewhere in the input (any lag, any dot). Under the first E-069 plan 80 % of hidden-dot targets broke this and 64 % of the loss was predict-the-mean.

2 • No copy targets

The anchor family scores hidden channels only (anchor_hidden_only); a target visible in the input is a copy, not a question.

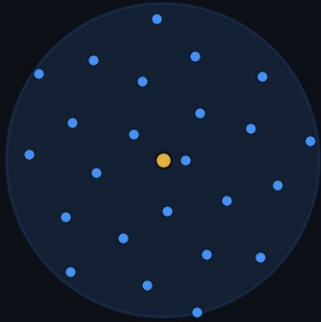
3 • Every family against its own bar

Per (task, family, lead) loss, each divided by its predict-the-mean score; a task is “learned” only where the ratio is below one at held-out anchors.

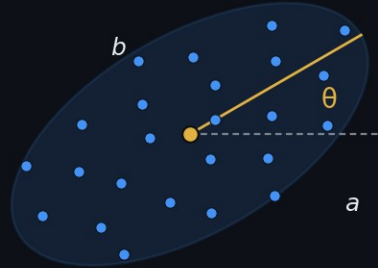
- The task is not told to the model by a token; the masks are visible (the mask_tok / miss_tok distinction stays), and the lead is in every query's coordinates — that is enough
- Loss weights: equal across tasks per example; the shares in the table set the mix, not the weights

Seven numbers per region shape the whole hourglass

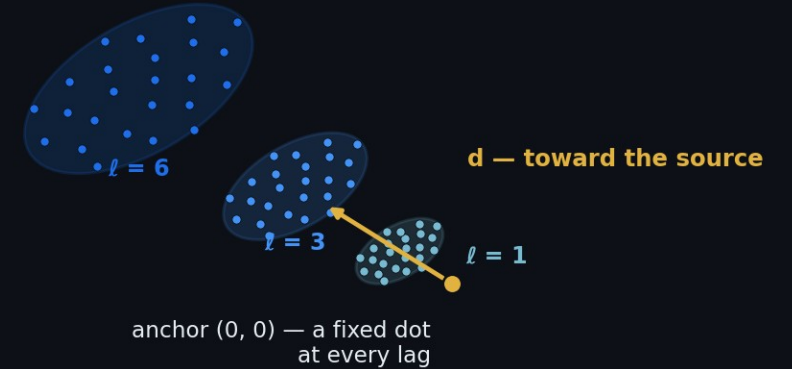
(a) canonical sunflower, unit disc
24 points, golden angle 137.508°



(b) rotated by $\theta = 30^\circ$, stretched to 2:1
semi-axes a, b



(c) the same pattern at three lags: shifted by $\ell \cdot d$, grown with ℓ



For each region and channel group, at lag ℓ (past, $\ell = 1 \dots 6$) the sample points are

$$p(\ell, i) = \text{anchor} + \ell \cdot \mathbf{d} + R(\theta) \cdot \text{diag}(a_0 + \ell \cdot s_a, b_0 + \ell \cdot s_b) \cdot \mathbf{u}_i$$

- $\mathbf{u}_i$: the canonical sunflower on the unit disc (Vogel spiral, golden angle), fixed, ~24 points — the same pattern at every lag
- $\mathbf{d}$ (2 numbers): the drift per pentad, in km — where the cone's centre moves as you go back in time; this is the arrow toward the source
- θ (1): orientation of the ellipse
- a_0, b_0 (2): the waist semi-axes, in km — the cut-off tip's size and elongation
- s_a, s_b (2): how fast each axis grows per pentad — spread
- **Future cone**: $p(-\ell)$ reflected, $c(+\ell) = -\ell \cdot d$, same axes at the same $|\ell|$; the anchor column (0, 0) is added at every lag regardless

Continuous positions, interpolated reads — the gradient reaches the geometry

Sample between the cells

A dot at a continuous position reads the field by **bilinear interpolation** over its four surrounding cells, weighting by the observed flags: $\text{value} = \sum w_i m_i v_i / \sum w_i m_i$, $\text{observed-fraction} = \sum w_i m_i$

The gradient exists

The interpolation weights are functions of the position, the position of the seven numbers, so $\partial \text{loss} / \partial (\mathbf{d}, \theta, a, b, s)$ exists — the deformable-convolution / spatial-transformer trick, applied to a gather instead of a convolution

Footprint, not point

Each dot averages over a Gaussian footprint whose width tracks the local dot spacing ($\sim r/\sqrt{n}$), read from a mip-map pyramid of the tensor — so a dot means “this area”, and the gradient is not confined to four cells

Same cost class as today

One gather per dot (4 cells, or one pyramid cell), on the fly, keyed exactly as E-069's examples are — plus the pyramid, built once per tensor

A 1° map per channel group, interpolated to the anchor

- A **1° grid** over the globe ($180 \times 360 = 64,800$ cells); the anchor's parameters are the bilinear blend of the four surrounding map cells — smooth by construction, no seams
- **1° because** the currents that carry the overturning are ~100 km wide (the Florida Current sits in an 80 km strait); a 10° cell would average the Gulf Stream with its recirculation and learn a drift near zero
- **Hierarchical prior:** each map is a coarse 10° map plus a 1° residual, the residual under L2 shrinkage and a Laplacian smoothness penalty — a data-poor cell (ice edge, coast) inherits its neighbourhood's arrow instead of fitting noise. Each 1° cell still sees $16 \text{ anchors} \times 3,142 \text{ pentads} \approx 50 \text{ k anchor-pentads}$ for seven numbers
- **No Argo ellipse.** The Argo channels are the k nearest profile tokens (E-071 §3, E-076) — placed where floats were, each carrying its own offset, age and depth. There is no elliptic shape to learn; the search radius stays as specified there. Depth-dependent sourcing (at the RAPID line the upper limb flows north, the Deep Western Boundary Current south) is left to attention over those tokens, which see depth and offset together
- **Optional, phase 2:** a first-harmonic seasonal term on the drift, $d(\tau) = d_0 + d_1 \cos \tau + d_2 \sin \tau$ — the Somali Current reverses with the monsoon

one map per channel group

ocean surface — currents, sea-surface height, sea-surface temperature, mixed layer

atmosphere — drift only at lags 0–1

land — no drift: $v = 0$, axes and angle only

≈ **1.4 M**

geometry parameters, nominal

64,800 cells $\times$ 3 groups $\times$ 7 numbers, about half of it live (ocean maps over ocean, land maps over land) — cheap, and no compute

Bounded by physics, tied to the fixed cone, started isotropic

Speed cap

$|\mathbf{d}| \leq v_{\text{design}} \cdot \Delta t = 5.4 \text{ m/s} \times 5 \text{ d} = 2,333 \text{ km per pentad (E-071's fastest mechanism} \times 1.5)$, via a tanh parameterisation — the learned cone lives inside the v2 envelope

Floors

$a_0, b_0 \geq \text{one cell (28 km)}$; $s_a, s_b \geq 0$; $a_0 + 6 s_a \leq v_{\text{design}} \cdot 6 \Delta t$

Prior

Coarse $10^\circ + 1^\circ$ residual, L2 shrinkage and Laplacian smoothness on the residual (slide 10) — one weight, recorded in the run's metrics

Mirror tie, by default

The future cone shares the seven numbers (reflected); an untied arm is the ablation, not the design

Stop-gradient on the target side

The loss reaches the geometry only through the **input** reads; the positions of targets are treated as constants

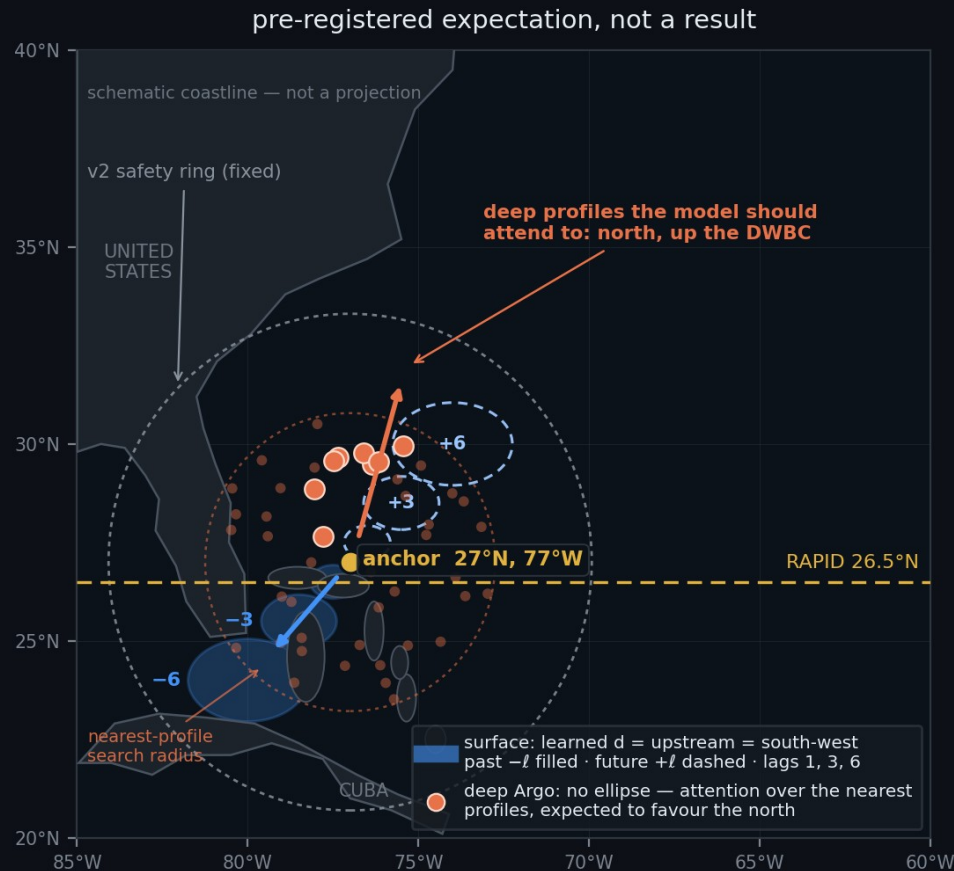
The v2 safety net stays

In addition to the learned ellipse, each lag keeps one fixed log-radial ring of 24 dots to the antipode — the cone can be directed without asserting that a driver could not have come from elsewhere

Init

$\mathbf{d} = 0, \theta = 0, a_0 = b_0 = 3 \text{ cells (aspect 0.71 applied)}$, s from E-069's 0.3 m/s — the fixed hourglass exactly, so step 0 of the learned arm is the control

One anchor on the western boundary: a learned surface cone, and deep profiles read from the other side



- **Surface group:** the learned d should point south-west (up the Florida Current, toward the Loop Current); the future cone leans north-east along the Gulf Stream
- **Deep Argo:** no ellipse to learn. The profile tokens within the search radius carry their own offset and depth; the model should put its attention on the deep (> 1000 dbar) profiles **north** of the anchor — up the Deep Western Boundary Current — and on the shallow ones to the south
- Neither is programmed: these are the **pre-registered expectations** — a geometry check for the surface, an attention check for the deep
- If the learned surface drift correlates with $-u_{\text{clim}}$ (the climatological surface current, reversed) over the world ocean, the cone learned “upstream” from the data alone

A learnable cone can learn to ask easy questions

The loophole: if target positions could move, the gradient would move them to smooth water — a lower loss that means nothing.

Fix 1 • stop-gradient on target positions

Slide 11. The geometry is trained only by “which inputs help”, never by “which targets are easy”.

Fix 2 • the mirror tie

The future cone cannot be tuned separately from the past cone.

Fix 3 • frozen evaluation geometry

Held-out numbers are always scored on the fixed hourglass, for every arm.

Residual second-order effect: as the input cone moves, the mirrored targets move with it. Monitored, not assumed away — report the target-variance under the learned cone vs the fixed cone at every eval; a drop > 10 % is a flag.

Tokens, gathers, parameters — the same regime as v2

	E-069 (measured)	this design (budget)
waist	42 patch tokens (14 ch × 3×3)	13 per-location tokens carrying all surface channels
past cone	706 dot tokens	6 lags × (1 + 24 ellipse + 24 ring) × 4 groups ≈ 1,176
Argo	192 (32 × 6, column only)	26 profile tokens (E-071 §3)
targets	42 + 84	256 dot queries per example, drawn from the future cone and the hidden waist / dots (as n_dot_queries today)
geometry parameters	0	≈ 1.4 M nominal, ~half live (slide 10); no compute
extra per-tensor artefact	—	the mip-map pyramid (~1/3 of the tensor, built once)

- Perceiver cost is linear in tokens: ~1.6× E-069's encoder time, in line with v2's own 2.1× estimate; the interpolated gather is 4 cells instead of 1
- Training cost per seed: E-069 seeds ran under one 4090-hour on the North Atlantic; budget 2–3 4090-hours per seed on the global tensor

Three arms, three seeds, four read-outs, one verdict rule

A0 • fixed

The hourglass with the v2 geometry (isotropic ellipses, ring, mirror), no learning — the control, and the eval geometry for everyone

A1 • learned

The seven numbers per region, initialised at A0

A2 • learned, primed

As A1, but **d** initialised at $-u_{\text{clim}} \cdot \Delta t$ (the climatological surface current, reversed) for the ocean surface group — tests whether the gradient can find upstream on its own (A1) or only keep it (A2)

Held fixed: same tensor (family 7.2, global), same 7 M codec, 20 k steps, frozen protocol (train ≤ 2020 ; 2008–09, 2016–17, 2021–24 held out), 3 seeds each.

R1 T1 future-cone loss per lead vs bar • **R2** the T1 – T2 gap (what history buys) • **R3** drift-vs-upstream correlation, world ocean, $|u_{\text{clim}}| > 0.2$ m/s

R4 the western-boundary depth check — surface drift south-west (geometry); attention mass on deep (> 1000 dbar) profile tokens north of the anchor, on shallow ones south (attention diagnostic)

R5 target-variance flag (slide 13)

Verdict: A1 adopted if it beats A0 on R1 at every lead ≤ 3 , paired at all three seeds, with R5 clean. $A2 > A1$ with $A1 \approx A0$ means the gradient cannot find upstream — then geometry is set from climatology, not learned. Neither $\rightarrow$ keep A0 and move to a dynamic (per-example) cone.

Five steps, each with its test

in ml/cone.py · ml/cone_sampler.py · ml/cone_codec.py · ml/train_cone.py

- 1 **Waist + future cone in the geometry (cone.py)** waist_dots, future_dots = reflected inner_dots; coverage_report extended with the mirror identity (future set = -past set) — pure numpy, CPU test
- 2 **The task mix (cone_codec.py::default_plan → plan_v3)** T1-T4 with shares and the five dropout patterns; test: every target's channel is visible somewhere in the input, for 10 k drawn examples, zero violations
- 3 **Continuous sampling (cone_sampler.py)** pyramid build + interpolated gather with observed-weighting; test: at integer positions with a full observed mask the read equals today's gather bit-for-bit
- 4 **The parameter maps (cone_codec.py::ConeGeometry)** 1° maps per group as coarse 10° + residual, tanh/softplus bounds, the shrinkage/smoothness prior, mirror, stop-gradient, init = A0; test: at init the dots equal step 1's fixed dots to < 0.1 cell; gradient-check on the seven numbers with finite differences
- 5 **Read-outs in the trainer (train_cone.py)** per-task bars, R3 drift maps exported as data/cone_drift_<group>.json for the app's Cones tab, R4 attention-by-depth-and-bearing at the western-boundary anchors, R5 flag; then the three arms as recipes f7l2-hourglass-{fixed,learned,primed}

Each step lands on main with its test before the next starts; step 3 is the only one that touches the data path and is gated on the bit-identity test.

Two settled, five still open — all reversible

Settled 15 Sep (Chris): 1° parameter maps with the coarse-plus-residual prior; **no Argo ellipses** — the nearest-profile tokens stay as E-071 §3 specifies them.

Future horizon

+6 pentads to mirror the past; or shorter (+1...+2 as E-069) with denser targets?

Task shares

0.35 / 0.15 / 0.25 / 0.25 — any of them a lever you want set differently?

Prior strength on the 1° residual

One weight, to be set by held-out loss on the training years — or fixed by hand for the first arm?

The v2 safety ring

Kept at every lag (24 tokens per lag per group); dropping it halves the past-cone tokens but restores v2's “too narrow” risk

Seasonal drift term

Deferred to phase 2; or in from the start (monsoon regions)?

Sources

the record this deck stands on

- **E-069 • Two stencils, one cone** — ml/plans/E069_cone_codec.md; results in ml/EXPERIMENTS.md#e-069 (five seeds, H1 refuted; what history buys)
- **E-069b masking-plan fix** — ml/cone_codec.py::default_plan (chan_drop_scope, anchor_hidden_only), ml/train_cone.py
- **E-071 • Cone v2** — ml/plans/E071_cone_v2.md (§3 profile tokens, §4 design speeds and log-radial dots, §4.5 the split in time, §6.4 per-location tokens)
- **E-076 • Family 8 nearest observations** — ml/plans/E076_family8_nearest_observations.md (the (value, Δx , Δy , Δt , n_R) token)
- **The geometry as built** — ml/cone.py (FAMILIES, reach_km, slots, inner_dots, outer_spiral, coverage_report)

and on

- **Deformable sampling:** Dai et al. 2017, Deformable Convolutional Networks; Jaderberg et al. 2015, Spatial Transformer Networks; Zhu et al. 2021, Deformable DETR
- **Currents:** E-071 §4.1's table (Somali 3.6 m/s, Gulf Stream ~2.5, DWBC 0.02–0.1 m/s); RAPID 26.5°N array for the limb directions